

Impact of early enteral nutrition in critically ill children: A systematic review and meta-analysis

Ana Cristina Assumpção Benjamin MD¹ | Humberto Magalhães Silva MD¹  |
 Raísa Sanches Uzun MD¹ | Andrea Maria Cordeiro Ventura MD, MSc² |
 Isabel de Siqueira Ferraz MD, MSc¹ | Roberto José Negrão Nogueira PhD³ |
 Tiago Henrique De Souza PhD¹ 

¹Pediatric Intensive Care Unit, Department of Pediatrics, Clinics Hospital of the State University of Campinas (UNICAMP), Campinas, São Paulo, Brazil

²Pediatric Intensive Care Unit, Department of Pediatrics, University Hospital of the University of São Paulo (USP), São Paulo, São Paulo, Brazil

³Department of Internal Medicine, School of Medical Sciences of the State University of Campinas (UNICAMP), Campinas, São Paulo, Brazil

Correspondence

Tiago Henrique De Souza, PhD, Department of Pediatrics, State University of Campinas (UNICAMP); 126, Tessália Vieira de Camargo St, 13083-887 Campinas, São Paulo, Brazil.
 Email: thsouza@unicamp.br

Abstract

Background: This study aims to evaluate the impact of early enteral nutrition (EEN) compared with late enteral nutrition on clinical outcomes in critically ill children.

Methods: PubMed, Embase, and the Cochrane Library were systematically searched until December 2024. The primary outcome was all-cause mortality, with secondary outcomes including duration of mechanical ventilation and length of stay in the pediatric intensive care unit (PICU) and hospital. The meta-analysis used a random-effects model with inverse variance weighting.

Results: Twenty-one studies (10,006 children) were included. Definitions of EEN varied across studies, ranging from 24 to 72 h. EEN was associated with decreased mortality in both randomized controlled trials (RCTs) (odds ratio [OR] = 0.64; 95% CI, 0.43–0.96; $P = 0.03$) and observational studies (OR = 0.38; 95% CI, 0.23–0.62; $P < 0.001$). A sensitivity analysis was conducted by combining studies with similar EEN definitions. EEN initiated within 24 h of PICU admission was not significantly associated with mortality (OR = 0.72; 95% CI, 0.43–1.20; $P = 0.21$). However, EEN within 48 h was significantly associated with reduced mortality (OR = 0.37; 95% CI, 0.25–0.56; $P < 0.001$). The certainty of evidence (Grading of Recommendations Assessment, Development and Evaluation) from RCTs was evaluated as low, whereas that from observational studies was evaluated as very low.

Conclusion: The evidence from this study suggests that EEN benefits critically ill children by reducing mortality and shortening hospital stays. However, the high risk of bias and very low certainty of the evidence highlight the need for further research.

KEYWORDS

critically ill children, early enteral nutrition, enteral nutrition, pediatric critical care medicine

Abbreviations: EEN, early enteral nutrition; IQR, interquartile range; LEN, late enteral nutrition; MD, mean difference; OR, odds ratio; PICU, pediatric intensive care unit; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; PROSPERO, International Prospective Register of Systematic Reviews; RCT, randomized controlled trial; RoB, risk of bias; TBI, traumatic brain injury.

INTRODUCTION

Critically ill children are at high risk for nutrition disorders, which are associated with increased morbidity and mortality.¹ On admission to the pediatric intensive care unit (PICU), high rates of malnutrition are observed, ranging from 7.2% to 64.7%, depending on the criteria used for nutrition assessment and the clinical and socioeconomic profile of the population.² These patients may experience a decline in their nutrition status because of the severe catabolism triggered by the organic response to stress, resulting in rapid depletion of nutrition stores, particularly muscle wasting.³ Therefore, prioritizing the development and implementation of optimal nutrition strategies is crucial to improve clinical outcomes for critically ill children.

The primary objective of critical care nutrition is to prevent and treat both macronutrient and micronutrient deficiencies, thereby averting malnutrition in well-nourished patients and halting further nutrition deterioration in those previously malnourished. Initiating enteral nutrition within 24–48 h of PICU admission is a recommended strategy to achieve this goal.^{4,5} Although there is no consensus on the definition of early enteral nutrition (EEN), its use has the potential to improve energy and protein delivery, which has been associated with reduced mortality.^{6,7} Beyond nutrition benefits, EEN may also contribute to maintaining gut integrity, preventing bacterial translocation, and modulating the systemic inflammatory response. However, most critically ill children may not receive EEN. A large pediatric study involving 5105 critically ill children reported that only 27.1% received enteral nutrition within the first 48 h of PICU admission.⁸ Although initiating and maintaining enteral nutrition in critically ill children presents several challenges, many of these hurdles are potentially avoidable.^{9,10}

Despite the theoretical advantages, there is a lack of evidence supporting the benefits of EEN in critically ill children, leading to differences in practices observed across various institutions.¹¹ Few studies have been conducted to evaluate the impact of EEN on clinical outcomes in the PICU.^{8,12–16} These studies have explored various end points, including mortality rates, duration of mechanical ventilation, incidence of infections, and length of PICU stay. Although some research suggests that EEN may offer significant benefits, others have reported inconclusive or conflicting results. Therefore, we conducted a systematic review and meta-analysis to comprehensively assess the impact of EEN on clinical outcomes in critically ill children.

METHODS

This systematic review and meta-analysis adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.¹⁷ The review protocol was registered in the International Prospective Register of Systematic Reviews database (CRD42024524086).

Search strategy

The initial search was conducted in March 2024 and subsequently updated in December of the same year. Searched literature databases included PubMed, Embase, and the Cochrane Library. Keywords for the search were “child,” “children,” “pediatric,” “paediatric,” “infant,” “infants,” “preschool child,” “preschool children,” “adolescent,” “adolescents,” “critical care,” “critically ill,” “intensive care,” and “critical illness” for population; “early nutrition,” “early enteral nutrition,” “early enteral feeding,” “late enteral nutrition,” “late enteral feeding,” “delayed enteral nutrition,” and “delayed enteral feeding” for exposure; and “mortality,” “infections,” “gastrointestinal complications,” “organ dysfunctions,” “mechanical ventilation duration,” “length of mechanical ventilation,” “length of hospital stay,” and “length of PICU stay,” for the outcomes. Other pertinent studies were sought through a manual search of the reference lists. When necessary, additional information was requested from the corresponding authors via email. The search was limited to studies published in English, Spanish, and Portuguese. No date restrictions were applied. Full details of the search strategies for all databases are available in the Supporting Information (page 2).

Study selection

Two authors (A.C.A.B. and T.H.d.S.) independently screened the titles and abstracts then the full texts. Final eligibility was determined according to predefined standard criteria. Disagreements were resolved through discussion and consensus, and, if needed, a third author (R.J.N.N.) was consulted.

Studies were included if they met the following criteria: (1) encompassed the majority of the population aged between 28 days and 18 years; (2) presented original data from interventional (randomized controlled trials [RCTs] or non-RCTs) or cohort studies; (3) compared EEN with delayed enteral nutrition in critically ill patients; and (4) reported at least one outcome of interest. The exclusion criteria were as follows: (1) investigated a population predominantly of neonates; (2) included those aged >18 years; and (3) were conference abstracts, editorials, case reports, case series, review articles, or observational studies without a control or comparator.

Outcome measures

The primary outcome was all-cause mortality, as defined in each study. Secondary outcomes included the achievement of nutrition goals, the incidence of healthcare-associated infections, the occurrence of gastrointestinal complications, the incidence of organ dysfunction, the duration of mechanical ventilation, and the lengths of stay in the PICU and hospital.

Data extraction

Two authors (A.C.A.B. and T.H.d.S.) independently extracted the following data from included studies: study design, sample size, participant characteristics, method for estimating nutrition requirement, and outcomes of interest. Discrepancies were resolved through discussion, and, if needed, a third author (R.J.N.N.) was consulted. No simplifications or assumptions were made.

Analysis of risk of bias and quality of evidence

The risk of bias (RoB) and quality of evidence was independently assessed by two authors (H.M.S. and R.S.U.). Discrepancies were resolved by discussions, revisiting the study, and, if needed, consultation with a third author (R.J.N.N.). The RoB in RCTs was evaluated using the Revised Cochrane RoB Tool (RoB 2).¹⁸ This tool includes five domains: "randomization process," "deviations from intended interventions," "missing outcome data," "measurement of the outcome," and "selection of the reported result." Each item was assigned one of the following three RoB judgments: "low RoB," "high RoB," or "some concerns." For nonrandomized studies, the RoB was evaluated by using The RoB In Nonrandomized Studies of Interventions tool.¹⁹ This tool contains seven domains, each evaluated for bias risk categorized as "low," "moderate," "serious," "critical," or "no information." The overall RoB in the included studies corresponds to the lowest judgment in any of the domains.

The quality of evidence for clinical outcomes was assessed according to recommendations of the GRADE working group.²⁰

Data synthesis and analysis

Descriptive statistics were reported using proportions and means with SDs or median and interquartile range unless otherwise noted. In the meta-analysis of skewed data, sample means and SDs were estimated following the method described by Wan et al.²¹

Meta-analyses were performed using random-effects models with inverse variance weighting in Review Manager software (version 5.3.5; Cochrane Collaboration). Dichotomous outcomes were compared using the odds ratio (OR), whereas continuous outcomes were compared using weighted mean differences (MDs), both with their 95% CIs. A two-sided $P < 0.05$ was considered statistically significant. If meta-analysis was hindered by exposure-outcome heterogeneity or an insufficient number of studies for a particular outcome or exposure, results were narratively presented in descriptive form.

Assessment of heterogeneity and reporting bias

To assess clinical heterogeneity, the studies were analyzed in subgroups stratified by exposure (ie, definition of EEN) and population characteristics. Statistical heterogeneity among studies was assessed

using both the Cochran Q statistic and I^2 statistic. Heterogeneity was considered to be statistically significant when $P < 0.05$ or $I^2 \geq 50\%$.¹⁷ The potential for publication bias was evaluated by visually examining funnel plots and conducting statistical tests, such as the Begg and Mazumdar adjusted rank correlation test and the Egger et al. regression asymmetry test.^{22,23}

RESULTS

Literature search

The search strategy found 10,737 studies. After removing 1171 duplicates, 9566 studies were screened by title and abstract. Of the 63 selected for full-text analysis, 21 met the inclusion criteria (Figure 1).

Characteristics of the included studies

This review included a total of 10,006 participants from four RCTs, 15 observational studies, and two secondary analyses of an RCT. Participants' median age ranged between 6 months and 13 years. Among the selected studies, five focused on patients with burn injuries, four on children with traumatic brain injuries, three on mechanically ventilated patients owing to respiratory failure, and eight on mixed surgical-clinical cases admitted to a general PICU. The main characteristics of the studies are shown in Tables 1 and 2.

EEN definitions

Definitions of EEN varied across the included studies (Tables 1 and 2). Nine studies assessed nutrition initiation within 48 h of hospital admission, seven studies within 24 h, and four studies within 72 h. The timeframe was <6 h in two studies.^{26,34} Most studies did not set a minimum nutrition intake that must be met within the specified period. Three studies defined EEN as reaching a minimum of 25% of the cumulative energy goal within 48 h,^{8,13,33} whereas one study required achieving 80% of the prescribed energy or protein goal within 72 h of PICU admission.¹⁵

Outcomes

Mortality

Four RCTs and 12 observational studies reported mortality and were included in the meta-analysis. Regardless of the varying exposure definitions adopted in the studies, EEN was associated with decreased mortality in both the subgroup of RCTs (OR = 0.64 [95% CI, 0.43–0.96]; $P = 0.03$; $n = 890$) and observational studies (OR = 0.38 [95% CI, 0.23–0.62]; $P < 0.001$; $n = 7409$) (Figure 2). Overall, EEN was associated

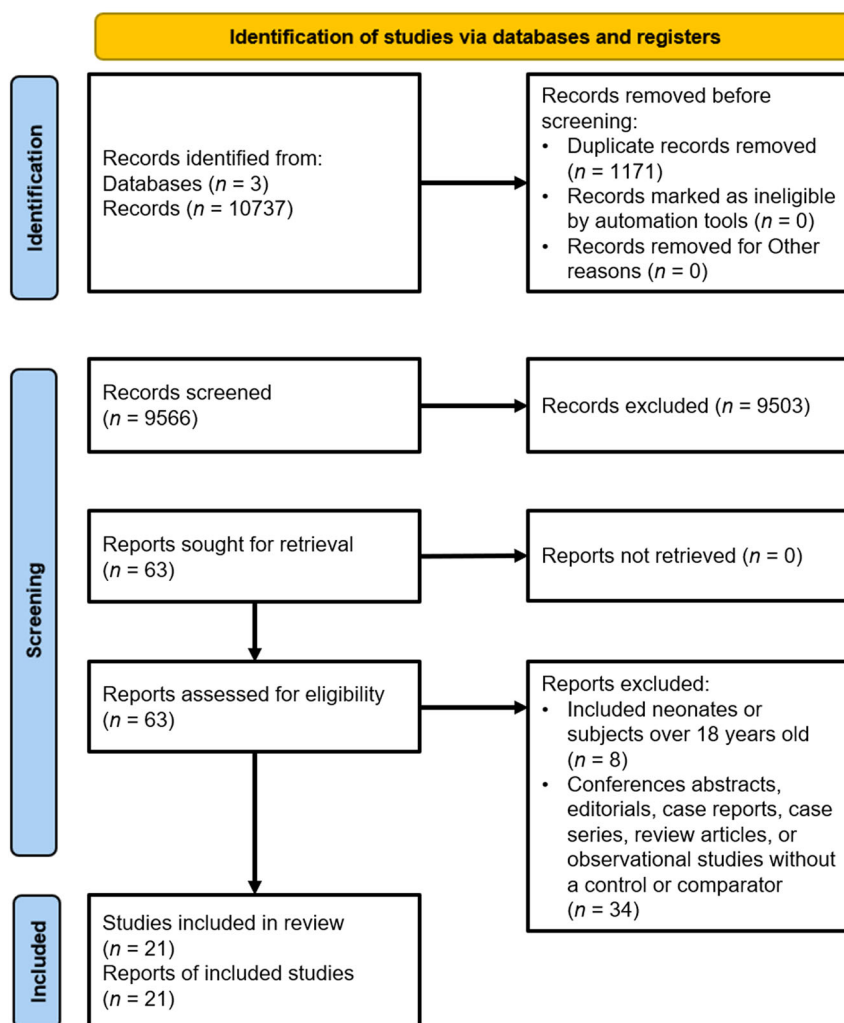


FIGURE 1 Flow chart of study selection and inclusion.

with reduced mortality, with a pooled OR of 0.46 [95% CI, 0.32–0.67; $P < 0.001$; $n = 8299$). Although no statistical heterogeneity was observed among RCTs ($I^2 = 0\%$), observational studies exhibited significant statistical heterogeneity ($I^2 = 50\%$). The sensitivity analysis identified the study by Sanches et al.²⁷ as the main source of heterogeneity in the subgroup of observational studies. On removing this study, EEN remained associated with lower mortality (OR = 0.41 [95% CI, 0.29–0.59]; $P < 0.001$), and no significant heterogeneity was observed among the remaining studies ($I^2 = 28\%$).

A sensitivity analysis was conducted by grouping studies sharing similar EEN definitions. EEN within 24 h of PICU admission was not significantly associated with mortality, with a pooled OR of 0.72 [95% CI, 0.43–1.20; $P < 0.21$; $I^2 = 20\%$; $n = 856$). However, when considering the 48-h timeframe, there was a significant association with reduced mortality (OR = 0.37 [95% CI, 0.25–0.56]; $P < 0.001$; $I^2 = 23\%$; $n = 6093$) (Figure 3). Additionally, analyses were conducted by grouping studies according to specific populations. The results can be found in the Supporting Information: Figure S1.

Only two of the studies conducted their analyses with adjustments for potential confounding factors. Mikhailov effect estimates were

adjusted for propensity score, Pediatric Index of Mortality 2 score, age, and center, whereas Powell et al. adjusted for age and the severity of pediatric acute respiratory distress syndrome. The meta-analysis, including these two studies, revealed no association of EEN with mortality (OR = 0.25 [95% CI, 0.04–1.60]; $P < 0.001$; $I^2 = 70\%$; $n = 5256$). Forest plot is available in the Supporting Information: Figure S2.

No evidence of publication bias was found in the analysis encompassing all studies (Egger test, $P = 0.987$; Begg test, $P = 0.805$), in the subgroup analysis of RCTs (Egger test, $P = 0.330$; Begg test, $P = 0.497$), or in the subgroup analysis of observational studies (Egger test, $P = 0.693$; Begg test, $P = 0.586$). Funnel plot can be found in the Supporting Information: Figure S3.

The RoB assessment was conducted for the primary outcome (mortality). Although two RCTs demonstrated an overall low RoB, two others were classified as having a high overall RoB owing to issues in the randomization process. Among the 17 observational studies, 12 were found to have a high overall RoB, whereas 5 were rated as having a moderate risk. The majority of these studies (70%) exhibited a high RoB in the confounding domain. The certainty of evidence (GRADE) from RCTs was evaluated as low, whereas that

TABLE 1 Primary characteristics of the included RCTs.

Study	Design and country	Timing for nutrition initiation	Population	Nutrition requirements estimation method	Outcomes
Prakash et al. ¹²	Open-label RCT; single center; India	EEN ≤ 6–24 h; LEN > 24 h	109 mixed clinical cases admitted to the general PICU: EEN = 57, LEN = 52	Hollyday and Segar formula	No differences in mortality, PICU length of stay, or other clinical outcomes No adverse events with EEN
Gottschlich et al. ²⁴	RCT; single center; USA	EEN ≤ 24 h; LEN > 48 h	72 children with ≥25% TBSA burn injuries: EEN = 36, mean age = 9.1 (SD = 0.7) y; LEN = 36, mean age = 9.6 (SD = 0.7) y	Indirect calorimetry	No differences in mortality, organ disfunctions, duration of mechanical ventilation, hospital length of stay, sepsis, diarrhea, or other clinical outcomes. The EEN group exhibited decreased energy deficits during postburn weeks 1 and 2 Serum insulin and triiodothyronine levels were elevated in the EEN group during postburn week 1, accompanied by a reduction in 3-methylhistidine output, indicating decreased protein breakdown
Venter et al. ²⁵	RCT; single center; South Africa	EEN ≤ 24 h; LEN > 48 h	21 children with ≥20% TBSA burn injuries: EEN = 11, median age 4.54 y; LEN = 10, median age 4.45 y	Indirect calorimetry	No differences in mortality, hospital length of stay, and loss of body weight The EEN group exhibited higher insulin levels ($P = 0.008$) and insulin-to-glucagon ratios, indicating an anabolic response
Khorasani et al. ²⁶	RCT; single center; Iran	EEN ≤ 3–6 h; LEN > 48 h	688 children with ≥10% TBSA burn injuries: EEN = 366, mean age = 5 (SD = 3.5) y; LEN = 322, mean age = 5 (SD = 3) y	Seashore method	The EEN group had reduced mortality (8.5% vs 12%), fewer hospital days (12.6 ± 1.3 vs 16.4 ± 3.7), and less body weight loss (3% vs 9%) compared with the LEN group ($P < 0.05$ for all)

Abbreviations: EEN, early enteral nutrition; LEN, late enteral nutrition; PICU, pediatric intensive care unit; RCT, randomized controlled trial; TBSA, total body surface area.

TABLE 2 Primary characteristics of the included observational studies.

Study	Design and country	Timing for nutrition initiation	Population	Nutrition requirements estimation method	Outcomes
Sánchez et al. ²⁷	Prospective observational study; single center; Spain	EEN = ≤24 h; LEN = >24 h of PICU admission (transpyloric)	526 children with critical illness; mean age = 23 (SD = 41) months; EEN (n = 202); LEN (n = 324)	-	No differences were noted in organ dysfunctions, vasoactive drug dosages, maximum energy delivery, nutrition duration, or mortality between groups. Abdominal distention incidence was lower in EEN group (3.5%) compared with the LEN group (7.8%; $P < 0.05$)
Taha et al. ²⁸	Retrospective study; single center; USA	Median (range) time for nutrition initiation of 1.49 days (0.02–11.88)	109 children with isolated TBI; median (range) age, 13 (8–18) y	-	The time to nutrition support initiation (enteral or parenteral nutrition) was positively associated with PICU length of stay ($r = 0.49$, $P < 0.01$) 92.3% of children received enteral nutrition
Mikhailov et al. ⁸	Retrospective multicenter study; 12 PICUs; USA	EEN was defined as 25% of cumulative energy goal ≤48 h of PICU admission	5105 critically ill children with PICU length of stay ≥96 h EEN (n = 1386) Median (IQR) age, 0.7 (0.3–2.9) y No EEN (n = 3719) Median (IQR) 4.0 (0.8–11.6) y	World Health Organization's equation	EEN was associated with lower mortality (adjusted OR, 0.51; 95% CI, 0.34–0.76; $P = 0.001$). No differences in duration of mechanical ventilation or length of stay were found
Vavilala et al. ²⁹	Retrospective multicenter study; 5 PICUs; USA	Early nutrition initiation <72 h of TBI	236 children with severe TBI Mean (SD) age, 8.0 (6.3) years	-	Early nutrition initiation (enteral or parenteral) was associated with mortality (adjusted hazard ratios, 0.06; 95% CI, 0.01–0.26)
Bagci et al. ¹⁴	Prospective observational study; multicenter; 9 PICUs; Turkey	EEN = ≤24 h; LEN = >24 h of PICU admission	95 critically ill children Mean (SD) age, 4 (4.7) y EEN (n = 45); LEN (n = 43)	World Health Organization's equation	There was no difference in mortality between the groups. Achievement of 25% of cumulative energy goal ≤48 h of PICU admission was associated with reduced 28-day mortality risk (adjusted OR, 0.06; 95% CI, 0.01–0.47; $P = 0.007$)
Haney et al. ¹⁵	Retrospective study; single center; USA	EEN = 80% of goal-prescribed energy or protein within 72 h of PICU admission	106 children with acute respiratory failure requiring mechanical ventilation EEN (n = 58) Median (IQR) age, 4 (1.0–17.0) months No EEN (n = 48) Median (IQR) age, 25.5 (2.0–144.5) months	World Health Organization's equation	EEN was associated with shorter PICU (adjusted $P = 0.001$) and hospital length of stay (adjusted $P < 0.001$) and delivery of higher percentage of goal kcal ($P < 0.001$) and protein ($P = 0.002$)

(Continues)

TABLE 2 (Continued)

Study	Design and country	Timing for nutrition initiation	Population	Nutrition requirements estimation method	Outcomes
Meinert et al. ¹⁶	Secondary analysis of an RCT; multicenter; 15 PICUs; USA, Australia, and New Zealand	Groups: (1) no nutrition over first 7 d; (2) <48 h after TBI; (3) 48 to <72 h after TBI; (4) 72–168 h after TBI	90 children with severe TBIBoth enteral and parenteral nutrition were considered in the analysis of nutrition support	–	Time to initiation of nutrition was associated with mortality ($P = 0.05$), with group 1 showing 22-fold higher odds of mortality compared with group 2. However, when adjusted for Hispanic ethnicity, this association did not reach statistical significance ($P = 0.10$)
Mikhailov et al. ¹³	Retrospective multicenter study; 2 PICUs; USA	EEEN was defined as 25% of cumulative energy goal ≤ 48 h of PICU admission	859 critically ill children: EEEN ($n = 171$); no EEEN ($n = 688$)	–	EEEN was associated with lower total (adjusted $P = 0.0006$) and daily (adjusted $P < 0.0001$) hospital charges
Balakrishnan et al. ³⁰	Retrospective multicenter study; 5 PICUs; USA	EEEN = ≤ 48 h; LEN = >48 h of PICU admission	416 children with TBI EEEN ($n = 347$) Median (IQR) age, 4.24 (0.75–12.42) y LEN ($n = 69$) Median (IQR) age, 6.00 (2.00–12.00) y	–	LEN was associated with lower Glasgow Coma Scale scores and higher Injury Severity Scores (both $P < 0.0001$), as well as worse functional status at PICU discharge ($P = 0.02$). After adjusting for confounders, EEEN showed no association with mortality or increased length of stay
Melro et al. ³¹	Prospective observational study; single center; Brazil	EEEN = ≤ 48 h; LEN = >48 h of PICU admission	71 children with respiratory failure: Median (IQR) age, 6 (2–13) months EEEN ($n = 49$) LEN ($n = 22$)	World Health Organization's equation Protein needs = 1.5 g/kg	The EEEN group received higher energy and protein content and presented improvement of nutrition adequacies than the LEN group ($P < 0.05$ for all) The time of EN initiation was not associated with the vasoactive drugs use, time of mechanical ventilation, PICU length of stay, and PICU mortality
Srinivasan et al. ³²	Secondary analysis of an RCT; multicenter; 31 PICUs; USA	EEEN = <48 h of RCT randomization	608 children with hyperglycemia requiring inotropic support and/or mechanical ventilation EEEN ($n = 331$); no EEEN ($n = 277$)	Schofield energy equations	EEEN was associated with lower mortality, reduced duration of mechanical ventilation, less organ dysfunction, and reduced PICU and hospital length of stay ($P < 0.05$ for all) The EEEN group reached target feeds faster than the no EEEN group. There were no differences between groups when stratified for average total daily energy intakes as percentages of their respective predicted energy goals. The EEEN group received a greater median percentage of total energy intake via the enteral route during study enrollment (84% vs 22%)

TABLE 2 (Continued)

Study	Design and country	Timing for nutrition initiation	Population	Nutrition requirements estimation method	Outcomes
Powell et al. ³³	Retrospective study; single center; USA	EEN = $\geq 25\%$ of cumulative energy goal within 48 h	151 infants and children with ARDS requiring mechanical ventilation: EEN (<i>n</i> = 41) Median (IQR) age, 1.85 (0.53–10.13) y No EEN (<i>n</i> = 110) Median (IQR) age, 6.00 (1.07–13.70) y	Schofield energy equations	EEN was associated with reduced mortality (adjusted OR, 0.071; 95% CI, 0.009–0.542; <i>P</i> = 0.011), shorter PICU length of stay (adjusted RR, 1.79; 95% CI, 1.25–2.55; <i>P</i> = 0.001), and more ventilator-free days (median, 13.8 vs 19.9 days; <i>P</i> = 0.017)
Shahi et al. ³⁴	Retrospective study; single center; USA	EEN = < 4 h; LEN = > 4 h of PICU admission	132 children with $\geq 10\%$ TBSA burn injuries: EEN (<i>n</i> = 63) Mean (SD) age, 4.35 (4.62) y LEN (<i>n</i> = 69) Mean (SD) age, 4.25 (3.99) y	Basal metabolic rate multiplied by a factor of 1.2 to 1.75 or indirect calorimetry, and protein needs were calculated at 1.5 to 4 g/kg	EEN group had lower rates of underfeeding during the first week (<i>P</i> = 0.014) and shorter PICU length of stay (<i>P</i> = 0.025). There was no difference in infectious complications, duration of mechanical ventilation, or hospital length of stay between groups
Solana et al. ³⁵	Prospective observational study; multicenter; 13 PICUs; Spain	EEN = ≤ 48 h; LEN = > 48 h of PICU admission	86 critically ill children: Median (IQR) age, 9.5 (3–42.3) months	Schofield energy equations or indirect calorimetry	EEN was associated with higher mean energy intake and maximum energy intake (<i>P</i> < 0.05 for both), but not with mean protein intake or maximum protein intake
Kurmis et al. ³⁶	Retrospective analysis of prospectively collected data; multicenter; Australia and New Zealand	EEN = ≤ 24 h; LEN = > 24 h of PICU admission	88 children with $\geq 15\%$ TBSA burn injuries: Median (IQR) age, 4.0 (1.0, 9.0) y EEN (<i>n</i> = 65); LEN (<i>n</i> = 11); no enteral nutrition (<i>n</i> = 10)	–	EEN was associated with reduced hospital length of stay (<i>P</i> = 0.029). There were no differences between the groups regarding the number of patients who were discharged to their home or usual residence
Solana et al. ³⁷	Secondary analysis of prospective observational study; multicenter; 12 PICUs; Spain	EEN vs LEN analyzed independently as ≤ 24 h vs > 24 h and ≤ 48 h vs > 48 h	68 critically ill children: Median (IQR) age, 8 (3–39) months 24h-EEN (<i>n</i> = 15); 24h-LEN (<i>n</i> = 53); 48h-EEN (<i>n</i> = 46); 48h-LEN (<i>n</i> = 22)	Schofield energy equations	24-h EEN vs 24-h LEN was associated with reduced duration of mechanical ventilation (<i>P</i> = 0.04) 48-h EEN vs 48-h LEN was associated with higher mean energy intake, faster attainment of maximum energy intake, and less constipation (<i>P</i> < 0.05 for all) There was no difference between EEN and LEN (24 or 48 h) regarding the percentage of children who reached the energy target
Zevallos et al. ³⁸	Retrospective study; single center; Peru	EEN = ≤ 72 h; LEN = > 72 h of PICU admission	370 critically ill children: Median (IQR) age, 19 (6–60) months EEN (<i>n</i> = 261); LEN (<i>n</i> = 109)	–	LEN was associated with longer hospital stays, extended PICU stays, and prolonged duration of mechanical ventilation compared with patients receiving EEN (<i>P</i> < 0.05 for all)

Abbreviations: ARDS, acute respiratory distress syndrome; EEN, early enteral nutrition; IQR, interquartile range; LEN, late enteral nutrition; OR, odds ratio; PICU, pediatric intensive care unit; RCT, randomized controlled trial; TBI, traumatic brain injury; TBSA, total body surface area.

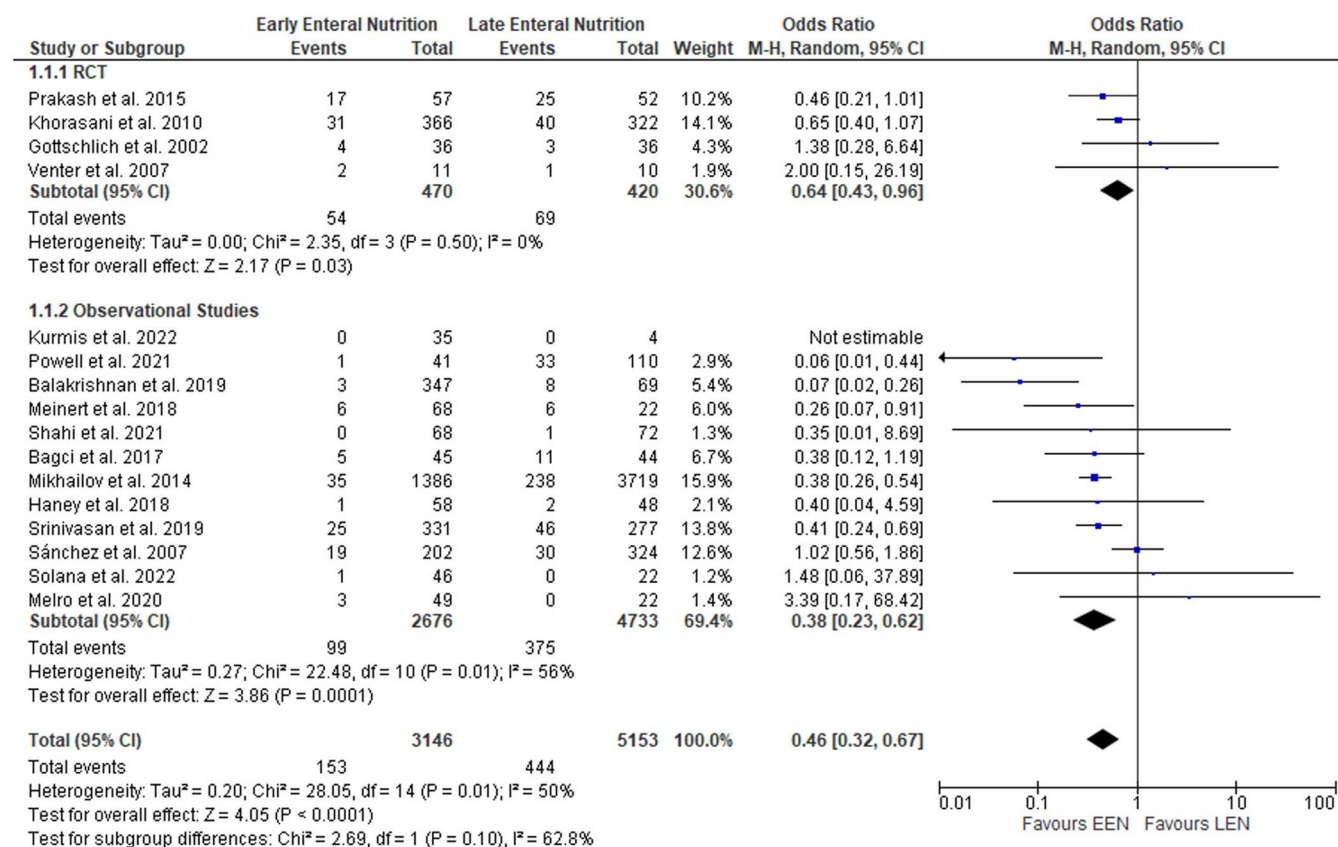


FIGURE 2 Association between EEN and mortality, stratified by specific study design. df, degrees of freedom; EEN, early enteral nutrition; LEN, late enteral nutrition; RCT, randomized controlled trial.

from observational studies was evaluated as very low. The comprehensive RoB analysis can be found in Tables 3 and 4, and the qualities of evidence for each outcome are presented in Table S1.

Duration of mechanical ventilation

Five studies reported outcomes on the duration of mechanical ventilation, with the meta-analysis revealing no significant difference between individuals who received EEN and those who received late enteral nutrition (LEN) (MD = 0.34 days [95% CI, -0.40 to 1.08]; $P = 0.25$; $I^2 = 26\%$). The forest plot can be found in the Supporting Information: Figure S4.

PICU and hospital length of stay

The meta-analysis of the six studies reporting PICU length of stay outcomes did not demonstrate a significant difference between EEN and LEN (MD = -0.93 h [95% CI, -2.57 to 0.70]; $P = 0.26$; $I^2 = 91\%$, $n = 6666$). Nonetheless, EEN was significantly associated with a reduced length of hospital stay (MD = -3.50 days [95% CI, -5.79 to -1.21]; $P = 0.003$; $I^2 = 86\%$, 7 studies, $n = 2597$). The forest plots of these analyses are available in the Supporting Information: Figures S5 and S6.

Other secondary outcomes

The assessment of healthcare-associated infections, gastrointestinal complications, and organ dysfunction was hindered by the insufficient reporting of outcomes in the included studies.

DISCUSSION

This systematic review and meta-analysis found that EEN is significantly associated with improved clinical outcomes in critically ill children, including reduced mortality and shorter hospital stays. To the best of our knowledge, this is the first meta-analysis evaluating the clinical impact of EEN compared with LEN in this population. However, given the very low certainty of the evidence, our findings should be interpreted with caution. The majority of the included studies had a high RoB and were primarily based on observational cohort studies. Only four studies assessed the impact of EEN in RCTs, three of which focused on specific populations of patients with major burns. Consequently, although we observed little statistical heterogeneity in the analyses, there was significant clinical heterogeneity among the studies.

Given the expectation of few RCTs comparing EEN with LEN in children, we decided to include nonrandomized studies in this

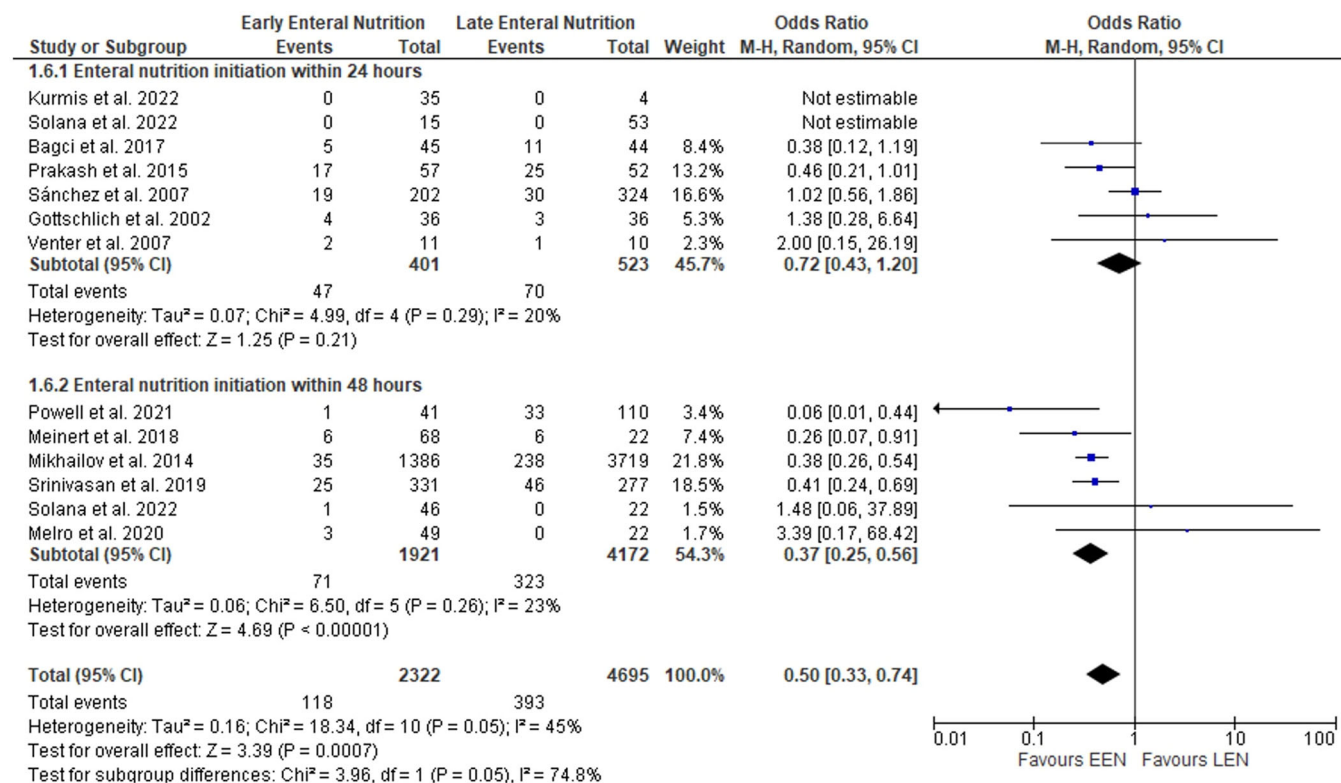


FIGURE 3 Association between EEN and mortality, stratified by timing of nutrition initiation. df, degrees of freedom; EEN, early enteral nutrition; LEN, late enteral nutrition.

TABLE 3 RoB-2 assessment for randomized trials.

Studies	RoB domains					Overall
	Randomization process	Deviations from the intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported result	
Prakash et al. ¹²	+	+	+	+	+	+
Gottschlich et al. ²⁴	+	+	+	+	+	+
Venter et al. ²⁵	×	+	+	+	+	×
Khorasani et al. ²⁶	×	+	+	+	+	×

Note: The judgment categories are as follows: + indicates a low RoB, - indicates some concerns, and × indicates a high RoB.

Abbreviation: RoB, risk of bias.

systematic review as well. However, evidence from nonrandomized studies is more susceptible to bias compared with RCTs. The RoB assessment identified significant concerns, especially regarding confounding bias. Unsurprisingly, this lack of adjustment for confounding variables in the included observational studies led to unadjusted estimates with the strongest association with reduced mortality found in this subgroup. Although there is a rationale for a causal relationship between EEN and improved outcomes, this evidence suggests only an association, not causality. Participants may have received LEN because they were more severely ill at PICU admission, potentially explaining their higher mortality rates. Therefore, our efforts to gather the most comprehensive evidence possible on the impact of EEN in children led to the inclusion of studies with a high

RoB. Consequently, the interpretation of some analyses requires careful consideration.

Evidence from the meta-analysis of RCTs was primarily derived from patients with major burn injuries. However, this specific population has unique nutrition needs owing to their prolonged hypermetabolic state and increased energy expenditure, which can exceed 250% of the basal metabolic rate.^{39,40} Burn patients often experience intense catabolism, where protein breakdown surpasses protein synthesis, leading to a negative net nitrogen balance and consequent muscle wasting.⁴⁰ Multiple factors contribute to this hypermetabolism, including a strong stress response and inflammatory reaction following severe burns, such as systemic inflammatory response syndrome, sepsis, septic shock, bloodstream infections, and the extent of the

TABLE 4 RoB In Nonrandomized Studies of Interventions-I RoB assessment for observational studies.

Studies	RoB domains							Overall
	Confounding	Selection of participants	Classification of intervention	Deviation from intended intervention	Missing data	Measurement of outcomes	Selection of reported results	
Sánchez et al. ²⁷	✗	+	+	-	-	+	-	✗
Taha et al. ²⁸	✗	+	-	-	+	+	-	✗
Mikhailov et al. ⁸	-	+	+	+	-	+	+	-
Vavilala et al. ²⁹	✗	+	+	-	-	-	-	✗
Bagci et al. ¹⁴	✗	+	-	-	-	-	-	✗
Haney et al. ¹⁵	✗	+	+	-	+	✗	-	✗
Meinert et al. ¹⁶	✗	-	-	-	+	+	-	✗
Mikhailov et al. ¹³	-	-	+	-	+	?	-	-
Blakrishnan et al. ³⁰	✗	-	+	-	+	-	-	✗
Melro et al. ³¹	-	-	-	-	-	+	-	-
Srinivasan et al. ³²	✗	+	+	+	-	?	-	✗
Powell et al. ³³	-	+	+	-	+	+	+	-
Shahi et al. ³⁴	✗	-	+	-	+	✗	?	✗
Solana et al. ³⁵	✗	+	+	-	+	✗	-	✗
Kurmis et al. ³⁶	✗	-	+	-	+	✗	-	✗
Solana et al. ³⁷	✗	-	+	-	+	✗	-	✗
Zevallos et al. ³⁸	-	-	+	+	-	+	-	-

Note: The judgment categories are as follows: + indicates a low RoB, - indicates a moderate RoB, ✗ indicates a serious RoB, and ? indicates no information.

Abbreviation: RoB, risk of bias.

burned area. Previous meta-analyses of studies evaluating adults with major burn injuries found that EEN was effective in reducing mortality, sepsis, gastrointestinal hemorrhage, and hospital length of stay.^{41,42} However, we did not find similar results in pediatric studies. Subgroup analysis of children with major burns showed no significant association between EEN and reduced mortality (OR = 0.71 [95% CI, 0.45–1.13]; $P = 0.15$; $I^2 = 0\%$; $n = 960$). Hence, the lack of definitive evidence regarding the benefits of EEN for children with major burns emphasizes the importance of conducting well-designed RCTs to fill this knowledge gap.

Currently, there is no consensus on the definition of EEN. Although the Society of Critical Care Medicine and the American Society for Parenteral and Enteral Nutrition (SCCM-ASPEN) advocate for initiating enteral nutrition within 24–48 h of PICU admission, the European Society of Pediatric and Neonatal Intensive Care (ESPNIC) guidelines recommend starting it within 24 h.^{4,5} Consequently, this systematic review identified a wide variety of EEN definitions adopted in the included studies. To contribute to the standardization of the EEN definition, we conducted a sensitivity analysis based on

the timing of enteral nutrition initiation. Interestingly, we found an association with reduced mortality in the subgroup of studies that initiated nutrition within 48 h but not in those that started within 24 h. It is worth noting that this analysis is limited by the observational nature of the studies. Nevertheless, there were few reports of complications related to EEN, indicating the safety of this nutrition strategy in patients with hemodynamic stability. Further research is needed in the pediatric population to determine the optimal timing of enteral nutrition as well as the minimum nutrition intake necessary for benefit outcomes. In addition, a critical gap in the literature is the scarcity of research on long-term outcomes such as growth, development, and quality of life after the ICU, which may be significantly affected by EEN.

Nonetheless, determining the optimal timing for initiating enteral nutrition in critically ill children poses significant challenges, as it involves a complex interplay of clinical factors that extend beyond a simple time-based framework. The impact of enteral nutrition cannot be effectively assessed by relying solely on the timing of PICU admission. Although EEN within 24 h may reduce mortality in

critically ill adults, findings from large RCTs suggest it provides no benefit and could even cause harm in specific populations, such as shock patients.^{43,44} Thus, a key question remains, "At what point is it safe to initiate enteral nutrition in critically ill children, particularly in those with borderline stability or specific conditions, such as shock or respiratory failure?" Addressing this question requires consideration of the child's hemodynamic stability, disease severity, nutrition needs, baseline nutrition status, metabolic and mineral disturbances, comorbidities, feeding tolerance, and the overall risk-benefit balance of EEN. Once safety criteria for initiating enteral nutrition are clearly defined, research can focus on assessing the impact of starting enteral nutrition at the ideal time; thus avoiding premature or delayed initiation. Although ethical concerns may prevent conducting interventional studies evaluating the impact of delayed enteral nutrition, other questions will still need to be addressed, such as the type of diet, amount provided, delivery method, and the timeframe for achieving nutrition targets.

The evidence supporting our findings is of very low certainty, largely owing to several important limitations. First, as previously mentioned, most included studies were observational, lacking large, multicenter RCTs. Second, the focus on patients with major burns in most RCTs limits the generalizability of the findings. Third, the majority of the included studies lacked adjustments for significant potential confounding factors, such as disease severity, nutrition status, nutrition requirements, and nutrition delivery. Therefore, their findings reflect only an association between EEN and improved outcomes without establishing causality. Finally, the varying definitions of EEN across studies introduced significant clinical heterogeneity into the analysis.

CONCLUSIONS

Based on very low certainty evidence, the findings of this meta-analysis suggest that EEN may positively impact critically ill children by reducing mortality rates and shortening hospital length of stay. However, the predominance of observational studies and the lack of large RCTs may introduce biases and confounders that could influence the observed associations. Therefore, further research with rigorous study designs is warranted to confirm these findings and establish causality.

AUTHOR CONTRIBUTIONS

Ana Cristina Assumpção Benjamin: Conceptualization; Investigation; Resources; Data curation; Writing—original draft; Visualization. **Humberto Magalhães Silva:** Data curation. **Raísa Sanches Uzun:** Data curation. **Andrea Maria Cordeiro Ventura:** Writing—review and editing. **Isabel de Siqueira Ferraz:** Writing—review and editing. **Roberto José Negrão Nogueira:** Writing—review and editing. **Tiago Henrique de Souza:** Conceptualization; Investigation; Writing—original draft; Formal analysis; Supervision.

ACKNOWLEDGMENTS

We thank Christopher Babbitt, MD (Miller Children's Hospital, Long Beach CA, USA) and María José Solana, PhD (Gregorio Marañón

University Hospital, Madrid, Spain) for providing additional data from their studies for this review.

CONFLICTS OF INTEREST STATEMENT

None declared.

ORCID

Humberto Magalhães Silva  <http://orcid.org/0009-0009-9671-7009>

Tiago Henrique De Souza  <http://orcid.org/0000-0001-6944-0221>

REFERENCES

- Ventura JC, Hauschild DB, Barbosa E, et al. Undernutrition at PICU admission is predictor of 60-day mortality and PICU length of stay in critically ill children. *J Acad Nutr Diet*. 2020;120(2):219-229. doi:10.1016/J.JAND.2019.06.250
- Moreno YMF, Ventura JC, Oliveira LDA, Silveira TT, Hauschild DB. Undernutrition in critically ill children. *Pediatr Med*. 2020;3:22. doi:10.21037/PM-20-66
- de Figueiredo RS, Nogueira RJN, Springer AMM, et al. Sarcopenia in critically ill children: a bedside assessment using point-of-care ultrasound and anthropometry. *Clin Nutr*. 2021;40(8):4871-4877. doi:10.1016/J.CLNU.2021.07.014
- Mehta NM, Skillman HE, Irving SY, et al. Guidelines for the provision and assessment of nutrition support therapy in the pediatric critically ill patient: society of critical care medicine and american society for parenteral and enteral nutrition. *Pediatr Crit Care Med*. 2017;18(7):675-715. doi:10.1097/PCC.0000000000001134
- Tume LN, Valla FV, Joosten K, et al. Nutritional support for children during critical illness: European Society of Pediatric and Neonatal Intensive Care (ESPNIC) metabolism, endocrine and nutrition section position statement and clinical recommendations. *Intensive Care Med*. 2020;46(3):411-425. doi:10.1007/S00134-019-05922-5
- Mehta NM, Bechard LJ, Cahill N, et al. Nutritional practices and their relationship to clinical outcomes in critically ill children—an international multicenter cohort study. *Crit Care Med*. 2012;40(7):2204-2211. doi:10.1097/CCM.0B013E31824E18A8
- Mehta NM, Bechard LJ, Zurakowski D, Duggan CP, Heyland DK. Adequate enteral protein intake is inversely associated with 60-d mortality in critically ill children: a multicenter, prospective, cohort study. *Am J Clin Nutr*. 2015;102(1):199-206. doi:10.3945/AJCN.114.104893
- Mikhailov TA, Kuhn EM, Manzi J, et al. Early enteral nutrition is associated with lower mortality in critically ill children. *JPEN J Parenter Enteral Nutr*. 2014;38(4):459-466. doi:10.1177/0148607113517903
- Mehta NM, McAleer D, Hamilton S, et al. Challenges to optimal enteral nutrition in a multidisciplinary pediatric intensive care unit. *JPEN J Parenter Enteral Nutr*. 2010;34(1):38-45. doi:10.1177/0148607109348065
- Panchal AK, Manzi J, Connolly S, et al. Safety of enteral feedings in critically ill children receiving vasoactive agents. *JPEN J Parenter Enteral Nutr*. 2016;40(2):236-241. doi:10.1177/0148607114546533
- Kerklaan D, Fizez T, Mehta NM, et al. Worldwide survey of nutritional practices in PICUs. *Pediatr Crit Care Med*. 2016;17(1):10-18. doi:10.1097/PCC.0000000000000542
- Prakash V, Parameswaran N, Biswal N. Early versus late enteral feeding in critically ill children: a randomized controlled trial. *Intensive Care Med*. 2016;42(3):481-482. doi:10.1007/S00134-015-4176-4
- Mikhailov TA, Gertz SJ, Kuhn EM, Scanlon MC, Rice TB, Goday PS. Early enteral nutrition is associated with significantly lower hospital charges in critically ill children. *JPEN J Parenter Enteral Nutr*. 2018;42(5):920-925. doi:10.1002/JPEN.1025

14. Bağcı S, Keleş E, Girgin F, et al. Early initiated feeding versus early reached target enteral nutrition in critically ill children: an observational study in paediatric intensive care units in Turkey. *J Paediatr Child Health*. 2018;54(5):480-486. doi:10.1111/JPC.13810
15. Haney A, Burritt E, Babbitt CJ. The impact of early enteral nutrition on pediatric acute respiratory failure. *Clin Nutr ESPEN*. 2018;26:42-46. doi:10.1016/J.CLNESP.2018.04.017
16. Meinert E, Bell MJ, Buttram S, et al. Initiating nutritional support before 72 hours is associated with favorable outcome after severe traumatic brain injury in children: a secondary analysis of a randomized, controlled trial of therapeutic hypothermia. *Pediatr Crit Care Med*. 2018;19(4):345-352. doi:10.1097/PCC.0000000000001471
17. Moher D, Liberati A, Tetzlaff J, Altman DG, et al. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *PLoS Med*. 2009;6(7):e1000097. doi:10.1371/journal.pmed.1000097
18. Sterne JAC, Savović J, Page MJ, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ*. 2019;366:4898. doi:10.1136/BMJ.L4898
19. Sterne JA, Hernán MA, Reeves BC, et al. ROBINS-I: a tool for assessing risk of bias in non-randomised studies of interventions. *BMJ*. 2016;355:4919. doi:10.1136/BMJ.L4919
20. Schünemann HJ, Oxman AD, Brozek J, et al. Grading quality of evidence and strength of recommendations for diagnostic tests and strategies. *BMJ*. 2008;336:1106-1110. doi:10.1136/BMJ.39500.677199.AE
21. Wan X, Wang W, Liu J, Tong T. Estimating the sample mean and standard deviation from the sample size, median, range and/or interquartile range. *BMC Med Res Methodol*. 2014;14:135. doi:10.1186/1471-2288-14-135
22. Egger M, Smith GD, Schneider M, Minder C. Bias in meta-analysis detected by a simple, graphical test. *BMJ*. 1997;315:629-634. doi:10.1136/bmj.315.7109.629
23. Begg CB, Mazumdar M. Operating characteristics of a rank correlation test for publication bias. *Biometrics*. 1994;50(4):1088-1101. doi:10.2307/2533446
24. Gottschlich MM, Jenkins ME, Mayes T, Khoury J, Kagan RJ, Warden GD. The 2002 clinical research award. *J Burn Care Rehabil*. 2002;23(6):401-415. doi:10.1097/01.BCR.0000036588.09166.F1
25. Venter M, Rode H, Sive A, Visser M. Enteral resuscitation and early enteral feeding in children with major burns—effect on McFarlane response to stress. *Burns*. 2007;33(4):464-471. doi:10.1016/J.BURNS.2006.08.008
26. Khorasani EN, Mansouri F. Effect of early enteral nutrition on morbidity and mortality in children with burns. *Burns*. 2010;36(7):1067-1071. doi:10.1016/J.BURNS.2009.12.005
27. Sánchez C, López-Herce J, Carrillo A, Mencía S, Vigil D. Early transpyloric enteral nutrition in critically ill children. *Nutrition*. 2007;23(1):16-22. doi:10.1016/J.NUT.2006.10.002
28. Taha AA, Badr L, Westlake C, Dee V, Mudit M, Tiras KL. Effect of early nutritional support on intensive care unit length of stay and neurological status at discharge in children with severe traumatic brain injury. *J Neurosci Nurs*. 2011;43(6):291-297. doi:10.1097/JNN.0B013E318234E9B2
29. Vavilala MS, Kernic MA, Wang J, et al. Acute care clinical indicators associated with discharge outcomes in children with severe traumatic brain injury. *Crit Care Med*. 2014;42(10):2258-2266. doi:10.1097/CCM.0000000000000507
30. Balakrishnan B, Flynn-O'Brien KT, Simpson PM, Dasgupta M, Hanson SJ. Enteral nutrition initiation in children admitted to pediatric intensive care units after traumatic brain injury. *Neurocrit Care*. 2019;30(1):193-200. doi:10.1007/S12028-018-0597-6
31. Melro EC, de Souza Lima AE, Missagia de Mattos Springer A, de Souza TH, Negrão Nogueira RJ. Protein intake deficiency in critically ill children with respiratory insufficiency: A call to action? *Clin Nutr ESPEN*. 2020;37:69-74. doi:10.1016/J.CLNESP.2020.03.018
32. Srinivasan V, Hasbani NR, Mehta NM, et al. Early enteral nutrition is associated with improved clinical outcomes in critically ill children: a secondary analysis of nutrition support in the heart and lung failure—pediatric insulin titration trial. *Pediatr Crit Care Med*. 2020;21(3):213-221. doi:10.1097/PCC.0000000000002135
33. Powell MBF, Rajapreyar P, Yan K, Sirinit J, Mikhailov TA. Nutrition practices and outcomes in patients with pediatric acute respiratory distress syndrome. *JPEN J Parenter Enteral Nutr*. 2022;46(6):1290-1297. doi:10.1002/JPEN.2320
34. Shahi N, Skillman HE, Phillips R, et al. Why delay? Early enteral nutrition in pediatric burn patients improves outcomes. *J Burn Care Res*. 2021;42(2):171-176. doi:10.1093/JBCR/IRAA129
35. Solana MJ, Manrique G, Fernández R, et al. Nutritional status and nutrition support in critically ill children in Spain: results of a multicentric study. *Nutrition*. 2021;84:110993. doi:10.1016/J.NUT.2020.110993
36. Kurmis R, Nicholls C, Singer Y, et al. An investigation of early enteral nutrition provision in major burn patients in Australia and New Zealand. *Nutr Diet*. 2022;79(5):582-589. doi:10.1111/1747-0080.12746
37. Solana MJ, Manrique G, Slocker M, et al. Early vs late enteral nutrition in pediatric intensive care unit: barriers, benefits, and complications. *Nutr Clin Pract*. 2023;38(2):442-448. doi:10.1002/NCP.10922
38. Zevallos MSC, Jara KSS, Ramírez CAC, Carreazo NY. Impact of early enteral nutrition on the hospital stay of pediatric patients undergoing mechanical ventilation. *J Pediatr Intens Care*. 2024;13(2):147-154. doi:10.1055/S-0042-1744296
39. Guo F, Zhou H, Wu J, et al. Prospective study on energy expenditure in patients with severe burns. *JPEN J Parenter Enteral Nutr*. 2021;45(1):146-151. doi:10.1002/JPEN.1819
40. Hart DW, Wolf SE, Mlcak R, et al. Persistence of muscle catabolism after severe burn. *Surgery*. 2000;128(2):312-319. doi:10.1067/MSY.2000.108059
41. Pu H, Doig GS, Heighes PT, Allingstrup MJ. Early enteral nutrition reduces mortality and improves other key outcomes in patients with major burn injury: a meta-analysis of randomized controlled trials. *Crit Care Med*. 2018;46(12):2036-2042. doi:10.1097/CCM.0000000000003445
42. Yang X, Li R, Zhai J, et al. Effects of early enteral nutrition in patients with severe burns: a systematic review. *Medicine (Baltimore)*. 2024;103:e37023. doi:10.1097/MD.00000000000037023
43. Reignier J, Plantefeve G, Mira JP, et al. Low versus standard calorie and protein feeding in ventilated adults with shock: a randomised, controlled, multicentre, open-label, parallel-group trial (NUTRIREA-3). *Lancet Respir Med*. 2023;11(7):602-612. doi:10.1016/S2213-2600(23)00092-9
44. Reignier J, Boisramé-Helms J, Brisard L, et al. Enteral versus parenteral early nutrition in ventilated adults with shock: a randomised, controlled, multicentre, open-label, parallel-group study (NUTRIREA-2). *Lancet*. 2018;391(10116):133-143. doi:10.1016/S0140-6736(17)32146-3

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Benjamin ACA, Silva HM, Uzun RS, et al. Impact of early enteral nutrition in critically ill children: a systematic review and meta-analysis. *J Parenter Enteral Nutr*. 2025;49:428-440. doi:10.1002/jpen.2738